1. Create a Custom VPC

🎯 **Objective:**

Set up a custom VPC with a /24 CIDR block, 3 subnets across different AZs, an IGW, and route configuration.

🛠 **Steps:**

1. **Create VPC:**
   - Name: Custom-VPC
   - CIDR block: 10.1.0.0/24
2. **Create Subnets** (in 3 different AZs):
   - Subnet-AZ1 → 10.1.0.0/26
   - Subnet-AZ2 → 10.1.0.64/26
   - Subnet-AZ3 → 10.1.0.128/26
3. **Create Internet Gateway (IGW):**
   - Name: Custom-IGW
   - Attach it to Custom-VPC
4. **Create a Route Table:**
   - Associate it with the 3 subnets.
   - Add a route: 0.0.0.0/0 → target: IGW
5. **Verify:**
   - Each subnet should show "Auto-assign Public IP: Yes" if you want internet access.

✅ **2. Test Cross-VPC EFS Access**

🎯 **Objective:**

Mount the same EFS volume from EC2 instances in different VPCs.

🛠 **Steps:**

1. **Create VPC-A and VPC-B** (each in the same region but different CIDRs, e.g., 10.0.0.0/24 and 10.2.0.0/24).
2. **Create an EFS file system:**
   - During setup, select "Enable access from other VPCs" (via EFS mount targets in both VPCs or Transit Gateway).
3. **Enable Cross-VPC Communication:**
   - Option 1: **VPC Peering**
     - Peer VPC-A and VPC-B.
     - Update route tables to allow traffic between CIDRs.

    - Option 2: **Transit Gateway** (if needed for large-scale setup)
4. **Launch EC2s in each VPC:**
    - Install NFS: sudo yum install -y amazon-efs-utils
    - Mount using: sudo mount -t efs fs-XXXX:/ /mnt/efs
5. **Security Group Rules:**
    - Ensure EFS SG allows NFS (port 2049) from both EC2 SGs.
6. **Test File Sync:**
    - Create a file from one EC2 and verify visibility from the other.

✅ **3. Troubleshoot Load Balancer Issues**

🎯 **Scenario:**

Simulate a broken environment by deleting the default VPC, then try to create a Classic Load Balancer (CLB). Restore the VPC and resolve.

🛠️ **Steps:**

1. **Delete Default VPC:**
    - Navigate to VPC → "Actions" → Delete Default VPC.
2. **Attempt to Create Classic Load Balancer:**
    - Go to EC2 → Load Balancers → Create CLB.
    - Observe error like:
        ❌ *"Classic Load Balancer requires a default VPC in this region"*
3. **Restore Default VPC:**
    - Open AWS Support → "Restore default VPC"
    - It takes a few minutes; refresh and verify VPC restored.
4. **Retry CLB Creation:**
    - This time it should succeed.

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4. Subnetting Practice

🔷 **Problem A: Divide 192.168.100.0/24 into 4 equal subnets**

Step-by-step solution:

- Original CIDR: /24 = 256 IPs
- To divide into 4 subnets → $2^2 = 4$ → Need 2 more bits
- New subnet mask: /26

- Each subnet has:
  - 2^(32-26) = 64 total IPs
  - 64 - 2 = 62 usable IPs

Resulting subnets:

S	Network Ad	Broadcast	Host Range
1	192.168.100	192.168.10	192.168.100.1 – 192.168.10
2	192.168.100	192.168.10	192.168.100.65 – 192.168.1
3	192.168.100	192.168.10	192.168.100.129 – 192.168.
4	192.168.100	192.168.10	192.168.100.193 – 192.168.

🔷 **Problem B: Calculate usable IPs for a /28 subnet**

- Subnet mask: /28 = 255.255.255.240
- Total IPs: 2^(32-28) = 16
- Usable IPs: 16 - 2 = 14

✅ **Answer: 14 usable IP addresses per /28 subnet**

✅ **5. Security Group Rules for EFS (Cross-VPC Access)**

🔷 **Minimum Required Rules**

Assuming EFS is mounted via **NFS (port 2049)** and peering is already in place or you're using a Transit Gateway.

✅ **Inbound Rules (on EFS SG)**

T	P	Por	Source	Descripti
		2049	EC2 SG / Peered V	Allow EC

You can use either:

- Specific **EC2 Security Group ID**
- Or **Peered VPC CIDR** (e.g., 10.1.0.0/24)

✅ **Outbound Rules (on EFS SG)**

T	P	Por	Des	Description
A	A	All	0.0.	Default (or restrict to VP

📄 Example JSON Format (EFS Security Group Inbound Rule)

```
json
CopyEdit
[
  {
    "IpProtocol": "tcp",
    "FromPort": 2049,
    "ToPort": 2049,
    "UserIdGroupPairs": [
      {
        "GroupId": "sg-0ecf12345ecf98765"   // EC2 SG ID
      }
    ]
  }
]
```

📄 Alternative: CIDR-Based JSON

```
json
CopyEdit
[
  {
    "IpProtocol": "tcp",
    "FromPort": 2049,
    "ToPort": 2049,
    "IpRanges": [
      {
        "CidrIp": "10.1.0.0/24"
      }
    ]
  }
]
```

Additional questions:

1. Subnetting in Real Deployments

✅ **Why Subnetting Matters Beyond Theory**

In AWS, subnets aren't just theoretical divisions—they control:

- **AZ-level redundancy**
- **Public vs Private resource separation**
- **Routing and security zones**

💡 **Real Use Case:**

You want to host a **web server, database, and admin VM**:

- Web Server → Public Subnet (Internet access)
- Database → Private Subnet (no IGW route)
- Admin VM → Bastion host in a Public Subnet

You might divide a /24 into:

- /26 for public subnet (web + bastion)
- /27 for private subnet (DB)
- /28 for admin or future scaling

🛑 **Mistake:**

Many beginners **allocate large CIDRs** unnecessarily and run into IP wastage or hit subnet count limits per VPC (200 max).

🔐 **2. Security Group Rules for EFS – Practical Notes**

💡 **Strategy:**

- EFS = Shared file system
- EC2 = Clients
- **NFS (2049)** must be open *only* to EC2s that need access (principle of least privilege)

🔐 **Minimum SG Rules Recap:**

- **Inbound (on EFS SG):**
  - TCP 2049 from EC2 SG or peered CIDR block
- **Outbound (on EC2 SG):**
  - Default outbound (all open) is usually enough

Insight:

Instead of using **CIDR ranges** (which can be error-prone), use **SG IDs** in rules:

```
json
CopyEdit
"UserIdGroupPairs": [
  {
    "GroupId": "sg-ec2-client-sg-id"
  }
]
```

Bonus Tip:

Use **VPC flow logs** to debug if traffic is reaching EFS:

- Filter by port 2049
- Look for REJECT entries if something's blocked

🌐 3. Custom VPC Design – Beyond Basics

What you gain from a custom VPC:

- **Granular control** over routing
- Control over **NAT, Firewall, DNS, DHCP**
- Multi-tier architecture layout (Web → App → DB)

🛠 Example Layout:

- 10.1.0.0/24 → VPC
  - 10.1.0.0/26 → Public Subnet (with IGW)
  - 10.1.0.64/26 → Private Subnet (with NAT)
  - 10.1.0.128/26 → DB Subnet (no outbound at all)

🎯 Interview Trick:

If asked, *"Why did you choose /26?"*

Say:

"Each /26 gives me 62 usable IPs, which is scalable for medium workloads, aligns well with AZ balancing, and avoids overprovisioning."

🛠 4. Classic Load Balancer Issue After Deleting Default VPC

❌ **What happens when you delete the default VPC?**

- AWS blocks certain **legacy services** like Classic Load Balancer
- Even **some marketplace AMIs fail** because they rely on default SGs or VPCs

Key Realization:

Default VPC isn't just a convenience—it's a dependency for backward compatibility features.

🛠 **Recovery Steps:**

1. Use AWS Support → Restore Default VPC
2. Wait for default subnets, IGW, and route tables to be auto-created
3. Retry the CLB setup

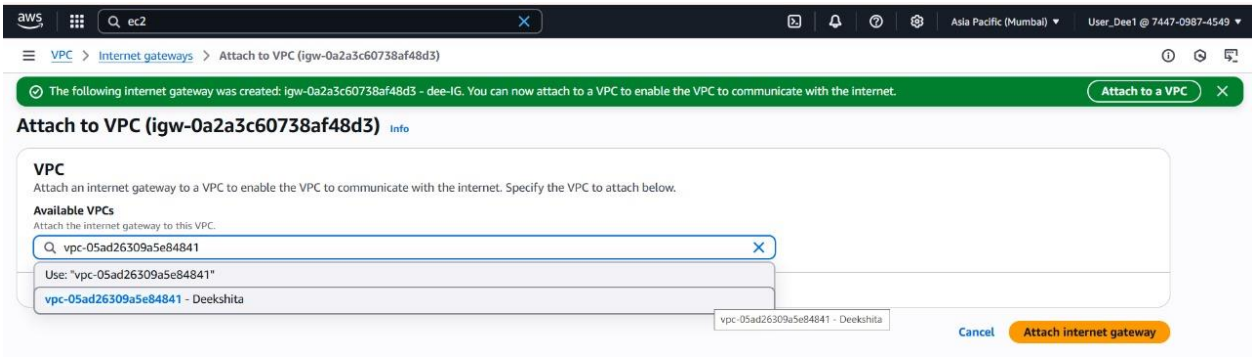
💬 **Tip**: In interviews, say:

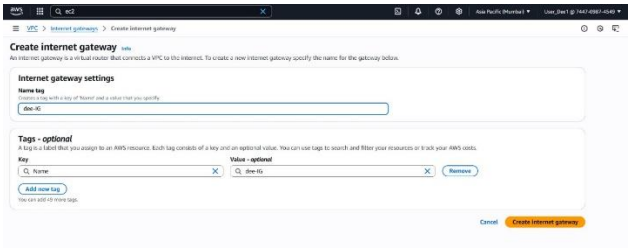
"While we generally use ALB or NLB, this was a legacy workload migration test that taught me how AWS services still depend on defaults."

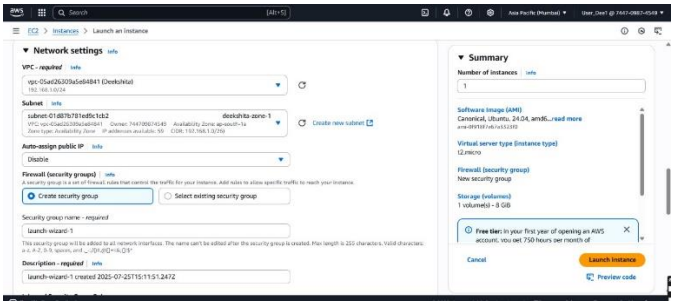
📊 **TL;DR Comparison Table**

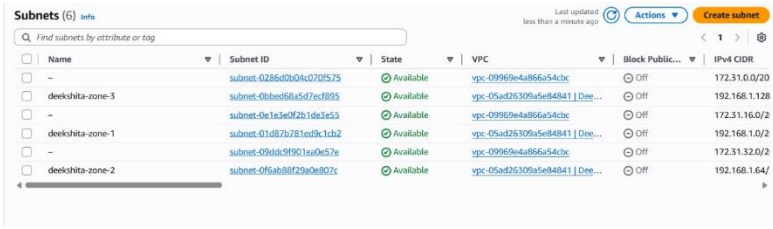
Topic	Key Skill Demonstrated	Real-World Challenge
Subnetting	CIDR math, IP planning	Efficient resource placement & scaling
EFS Security Groups	NFS networking, least privilege	Data breach prevention, cross-VPC access
Custom VPC	VPC architecture, routing	Isolated tiers, HA design
CLB Troubleshooting	AWS limits & legacy dependencies	Migration of old workloads

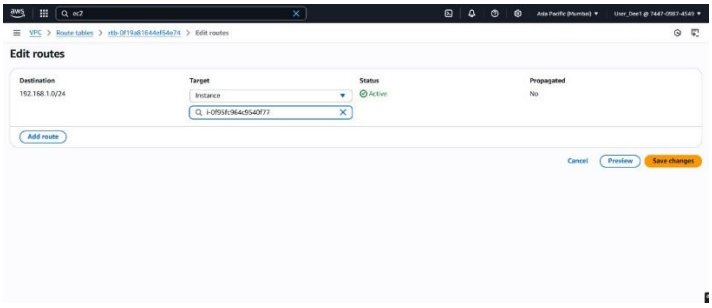
References:

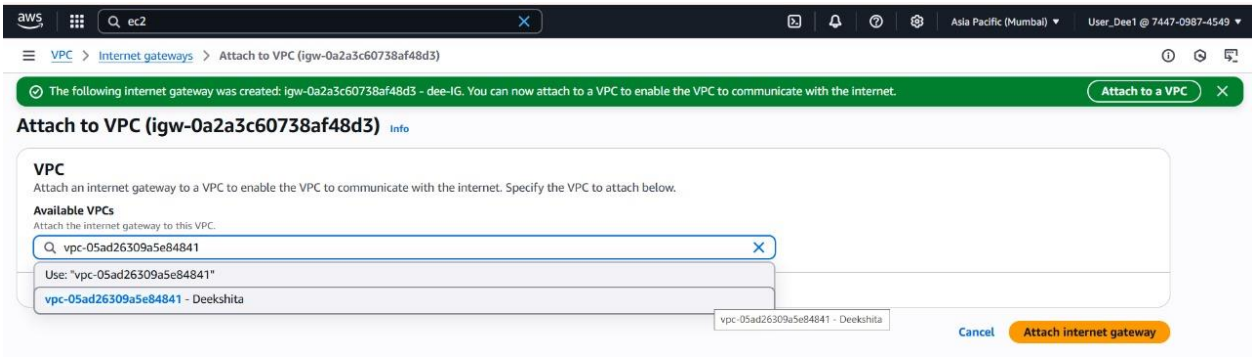



Create internet gateway
Internet gateway settings
Name tag
dee-IG
Tags - optional
Key
Value - optional
Name
dee-IG
Remove
Add new tag
Cancel
Create internet gateway


Network settings
VPC - required
vpc-05ad26309a5e84841 (Deekshita)
Subnet
subnet-01d87b781ed9c1cb2
deekshita-zone-1
Create new subnet
Auto-assign public IP
Disable
Firewall (security groups)
Create security group
Select existing security group
Security group name - required
launch wizard-1
Description - required
launch-wizard-1 created 2025-07-25T15:11:51.2472
Summary
Number of instances
1
Software image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of
Cancel
Launch instance
Preview code


Subnets (6)
Actions
Create subnet
Find subnets by attribute or tag
Name
Subnet ID
State
VPC
Block Public...
IPv4 CIDR
subnet-0286d0b04c070f575
Available
vpc-09969e4a866a54cbc
Off
172.31.0.0/20
deekshita-zone-3
subnet-0bbed68a5d7ecf895
vpc-05ad26309a5e84841 | Dee...
192.168.1.128
subnet-0e1e3e0f2b1de3e55
172.31.16.0/2
deekshita-zone-1
subnet-01d87b781ed9c1cb2
192.168.1.0/2
subnet-09ddc9f901ea0e57e
172.31.32.0/2
deekshita-zone-2
subnet-0f6ab88f29a0e807c
192.168.1.64/


Edit routes
Destination
Target
Status
Propagated
192.168.1.0/24
Instance
i-0f95fc964c9540f77
Active
No
Add route
Cancel
Preview
Save changes


ec2
Asia Pacific (Mumbai)
User_Dee1 @ 7447-0987-4549
VPC > Internet gateways > Attach to VPC (igw-0a2a3c60738af48d3)
The following internet gateway was created: igw-0a2a3c60738af48d3 - dee-IG. You can now attach to a VPC to enable the VPC to communicate with the internet.
Attach to a VPC
Attach to VPC (igw-0a2a3c60738af48d3)
Info
VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.
Available VPCs
Attach the internet gateway to this VPC.
vpc-05ad26309a5e84841
Use: "vpc-05ad26309a5e84841"
vpc-05ad26309a5e84841 - Deekshita
vpc-05ad26309a5e84841 - Deekshita
Cancel
Attach internet gateway